

Marker-Based and Marker-Less Motion Capture for Measuring Elbow Kinematics During a Drinking Task

Prem Khanvilkar, Christophorus Justin Setiawan, Farrel Aryo Wahyudi

Biomedical Engineering, The University of British Columbia

BMEG 230 Lab 2 · Lab Section L1B

Abstract

Kinematic motion capture quantifies how the body moves, and both marker-based and marker-less systems are used for this. This study applied both approaches to a simple drinking task performed with each arm. Reflective markers on the shoulder, elbow and wrist were tracked in video with Tracker to obtain the elbow angle, angular velocity and wrist tangential velocity, and a depth-sensor system (Nuitrack) was compared with a goniometer for two static elbow poses. After 1.5 Hz low-pass filtering, the left elbow showed a larger range of motion than the right (0.4851 vs. 0.4121 rad), whereas the right arm reached higher peak angular velocity (0.5512 vs. 0.4879 rad/s) and peak wrist tangential velocity (0.1626 vs. 0.1439 m/s), with between-arm differences of 12–16%. The depth sensor agreed with the goniometer within 0.25% at 80° but differed by 10.22% at 45°. With a single trial per arm and two static poses, the results are exploratory.

Keywords: *motion capture, marker-based tracking, marker-less tracking, elbow kinematics, angular velocity, depth sensor*

1. Introduction

Measuring how joints move is central to biomechanics. Kinematic data such as joint angles, range of motion (ROM) and velocities are used to assess movement quality, track recovery in rehabilitation, analyse athletic technique and design assistive devices. Recent advances in sensing have made these measurements both more precise and more practical to collect.

Marker-based tracking is the traditional approach. Physical markers are placed on anatomical landmarks and their positions are followed in video or by dedicated cameras, either manually or automatically, to reconstruct joint motion. It gives direct, time-resolved measurements, but placing markers takes time and marker shift or misplacement introduces error. Marker-less systems instead use depth-sensing cameras and skeleton-tracking software to estimate joint positions without anything attached to the body. They are quicker to set up and capture three-dimensional information, but their accuracy depends on camera alignment, lighting and how clearly the joints are visible.

This experiment applied both approaches to elbow motion. The objective was to characterise elbow angle, angular velocity and wrist tangential velocity in the left and right arms with a marker-based system, and to assess the accuracy of a marker-less depth-sensor system against a goniometer.

2. Methods

2.1 Marker-Based Motion Capture

The motion of interest was reaching for a water bottle, lifting it to drink, and returning it to its starting position. The task was recorded in side-view video for the left and right arms in separate trials. Reflective markers were placed on the shoulder, elbow and wrist to track the arm (Figure 1). The video clips were

imported into Tracker, where the markers were tracked and the kinematic analysis was performed (Figure 2).

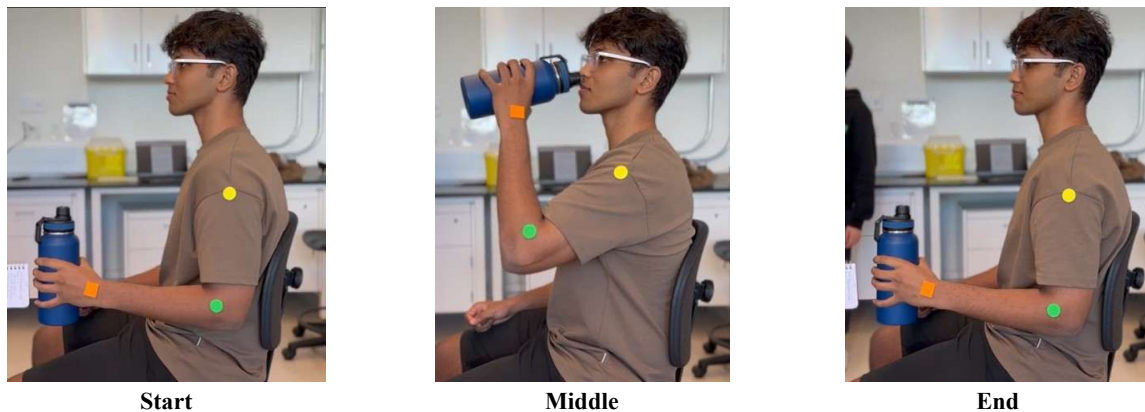


Figure 1. Progression of the drinking task with markers on the shoulder (yellow), elbow (green) and wrist (orange).

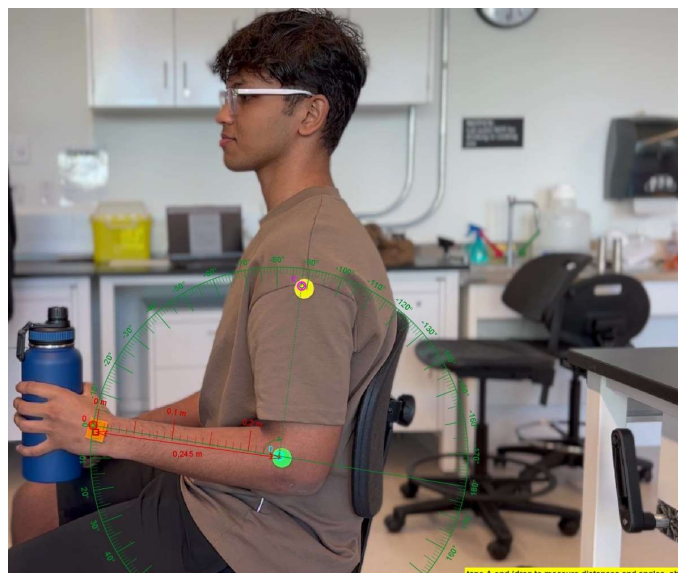


Figure 2. Tracker view showing the tracked shoulder, elbow and wrist markers, the calibration tape and the protractor used to measure the elbow angle (left-arm trial).

2.2 Marker-Less Motion Capture

The marker-less system used a depth-sensing camera with NuiTrack skeleton tracking to capture two static elbow poses, and screenshots of each pose were saved. A goniometer measured the real elbow angle at the same time. The Tracker protractor tool was then applied to each screenshot to estimate the elbow angle more precisely, and the estimates were compared with the goniometer readings.

2.3 Data Analysis

Kinematic data were exported from Tracker and processed in MATLAB to obtain the elbow joint angle, elbow angular velocity and wrist tangential velocity. The analysis assumed that all motion occurs in the sagittal plane, that the forearm behaves as a rigid body of fixed, measured length, that camera alignment errors are negligible, and that consistent marker placement and filtering limit manual tracking variability.

Elbow angle. The elbow angle was measured with the Tracker protractor tool from the shoulder, elbow and wrist markers and plotted against time for both arms to show the ROM, defined as the maximum minus the minimum of the filtered angle.

Filtering. To choose a cutoff, the fast Fourier transform (FFT) of the left and right angle signals was examined. A 4th-order Butterworth low-pass filter with a 1.5 Hz cutoff was then applied, designed for a 30 Hz sampling rate and run forward and backward (MATLAB filtfilt) to avoid phase shift. The low cutoff removes high-frequency tracking noise while keeping the low-frequency content that represents the range of motion. Derived signals were computed from the filtered angle so that they were also smooth.

Angular and tangential velocity. Angular velocity was calculated as the numerical derivative of the filtered elbow angle with respect to time. Wrist tangential velocity was estimated from the angular velocity and the measured forearm length, $r = 0.295$ m:

$$v_{wrist}(t) = r \cdot \omega(t)$$

Peak values were taken as the maximum of each signal in MATLAB. Between-arm differences in Table 1 are reported as the difference divided by the mean of the two arms:

$$\% \text{ difference} = |L - R| / ((L + R) / 2) \times 100\%$$

Marker-less accuracy. The accuracy of the depth-sensor angles was expressed as a percentage difference from the goniometer reference:

$$\% \text{ difference} = |\theta_{tracker} - \theta_{goniometer}| / \theta_{goniometer} \times 100\%$$

3. Results

3.1 Marker-Based System

Figure 3 shows the filtered elbow angle for both arms, Figure 4 the angular velocity with the peak of each arm marked, and Figure 5 the wrist tangential velocity. Table 1 summarises the ROM and peak values.

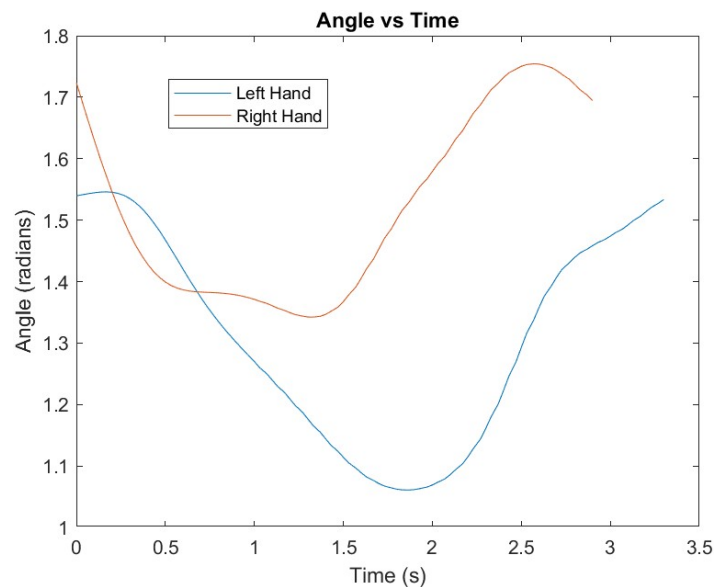


Figure 3. Filtered elbow angle against time for the left and right arms.

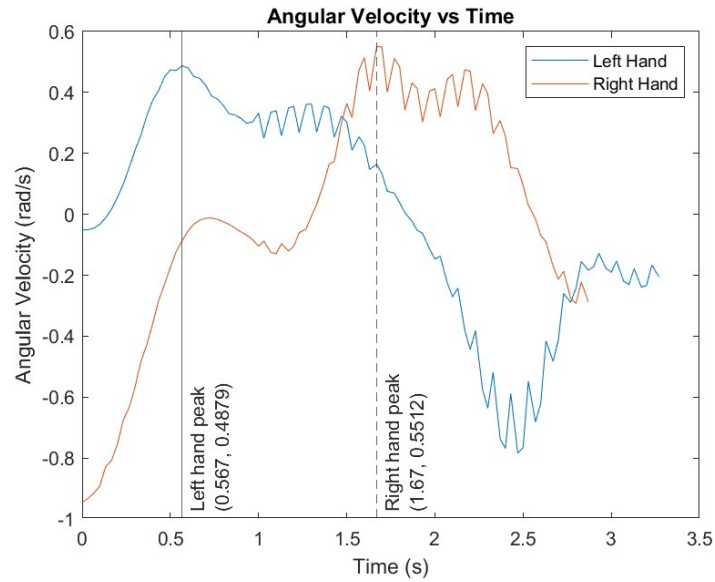


Figure 4. Filtered elbow angular velocity against time. Vertical lines mark each arm's peak: left 0.4879 rad/s at 0.567 s, right 0.5512 rad/s at 1.67 s.

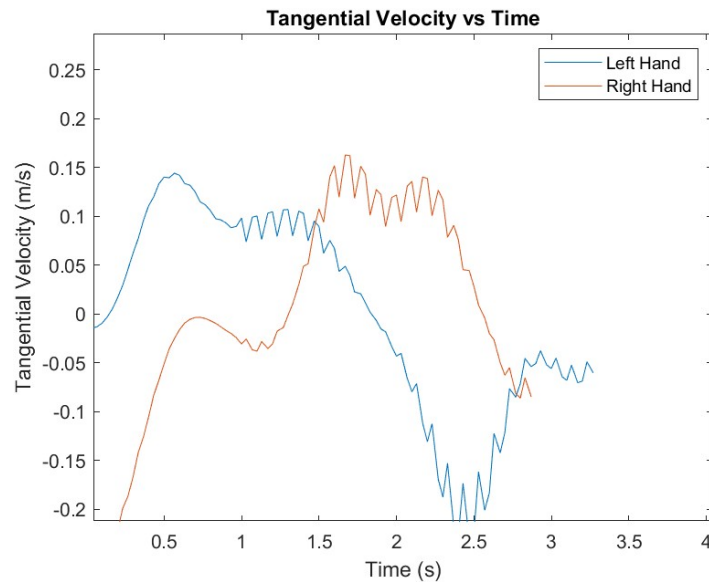


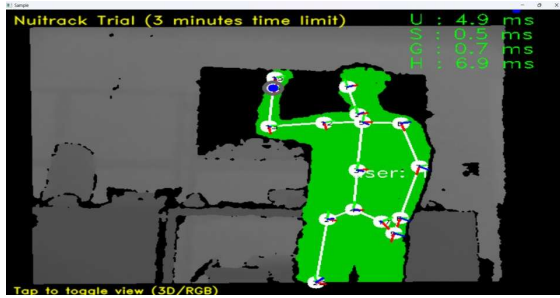
Figure 5. Filtered wrist tangential velocity against time for the left and right arms.

Table 1. Marker-based results. Percentage difference is between arms, relative to their mean.

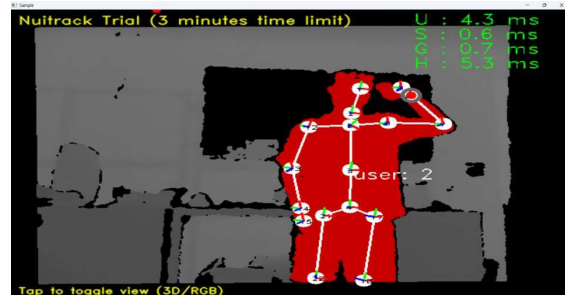
Parameter	Left arm	Right arm	% difference
Elbow ROM (rad)	0.4851 (27.8°)	0.4121 (23.6°)	16.27
Peak angular velocity (rad/s)	0.4879	0.5512	12.18
Peak tangential velocity (m/s)	0.1439	0.1626	12.20

3.2 Marker-Less System

Figure 6 shows the two static poses captured by the depth sensor. Table 2 compares the goniometer angles with the angles estimated in Tracker from the screenshots.



(a) Pose 1: right arm, 80°



(b) Pose 2: left arm, 45°

Figure 6. Static elbow poses captured with the depth sensor and NuiTrack skeleton tracking.

Table 2. Comparison of goniometer and Tracker-estimated elbow angles.

Pose	Goniometer (°)	Tracker (°)	Error (°)	% difference
1 (80°)	80	80.2	0.2	0.25
2 (45°)	45	40.4	4.6	10.22

4. Discussion

4.1 Left–Right Differences

The left arm covered a larger ROM than the right (0.4851 vs. 0.4121 rad, about 27.8° vs. 23.6°), indicating greater joint excursion. The right arm, however, reached a higher peak angular velocity and a higher peak wrist tangential velocity, each about 12% larger. Overall the two arms differed by 12–16%. Possible explanations include hand dominance, differences in neuromuscular control, small inconsistencies in marker placement and variation in how the motion was performed. With one trial per arm these factors cannot be separated, and the recordings also differ in duration and phase (the left peak occurs at 0.567 s and the right at 1.67 s), so the differences should be treated as observations rather than evidence of a true asymmetry.

4.2 Marker-Less Accuracy

The depth sensor agreed closely with the goniometer for the 80° pose (0.25%, or 0.2°) but showed a larger difference for the 45° pose (10.22%, or 4.6°). Part of this gap is arithmetic, since the same absolute error gives a larger percentage at a smaller angle, but it may also reflect reduced reliability at smaller joint angles and imperfect camera alignment. The comparison also combines several sources of error, because the angle was estimated by applying a protractor to a screenshot of the skeleton rather than read directly from the sensor. With only two poses, it suggests that the system gives reasonable estimates under suitable configurations rather than characterising its accuracy.

The main advantages of the marker-less system are that it captures three-dimensional spatial data without physical markers and does so quickly. Its main limitations are that accuracy drops when joints are not visible to the camera and that lighting and background conditions add noise to the measurements.

4.3 Limitations of the Marker-Based Approach

- Marker placement is a source of human error, particularly when markers shift during the motion.
- The 2D sagittal-plane analysis ignores out-of-plane movement and may oversimplify a three-dimensional motion.

- Each arm was recorded once, so trial-to-trial variability is unknown.
- Peak angular velocity was defined as the maximum of the signed velocity. Larger magnitudes occur in the opposite direction (about -0.8 rad/s for the left arm), and the right-arm trace starts at a large negative value where filter edge effects may be present, so a magnitude-based peak would be higher.
- Differentiating the angle amplifies noise, which is visible as small oscillations in the velocity traces even after filtering.

4.4 Future Work

Repeating trials, recording more participants and measuring joint acceleration and movement-smoothness indices would give deeper insight into motor control and the quality of the motion. Adding more static poses across a range of angles would show how depth-sensor accuracy varies with joint angle.

5. Conclusion

This lab compared marker-based and marker-less tracking of elbow motion. The marker-based analysis found differences of 12–16% between the arms: the left elbow had the larger ROM and the right arm the higher peak angular and wrist tangential velocities. The marker-less depth sensor matched the goniometer closely at 80° (0.25%) but less well at 45° (10.22%). Marker-based tracking is better suited to detailed, time-resolved kinematic analysis, while marker-less tracking is a practical alternative for quick assessments of posture and angle. Both findings rest on single trials and few poses.

Author Contributions

P. Khanvilkar: results, MATLAB code, appendix and data analysis. F. A. Wahyudi: data analysis, methods and background. C. J. Setiawan: discussion, conclusion and methods.

Acknowledgement of AI Assistance

ChatGPT was used to verify and refine the MATLAB code, with prompts asking for the error in the range-of-motion code.